Artifact Description:
SkyHOSTop: Cost-Aware Multi-Cloud Streaming Overlay Optimization

I. Overview

This document describes the research artifacts accompanying the paper *"SkyHOSTop: Cost-Aware Multi-Cloud Streaming Overlay Optimization."* The artifacts include measurement datasets, the MILP solver implementation, evaluation scripts, baseline implementations, pre-computed results, and figure-generation scripts. All materials are provided to enable reviewers and researchers to verify, reproduce, and extend the results reported in the paper.

II. Access & Links

All artifacts are hosted in a public, anonymized repository:

https://anonymous.4open.science/r/SkyHOSTop-evaluation/

The repository contains:

Path	**Description**
`data/mc_throughput_real.csv`	Per-VM throughput (Gbps), iperf3
`data/mc_latency_real.csv`	Inter-region RTT (ms), ping
`data/mc_cost.csv`	Egress cost ($/GB), provider pricing
`solver/solver.py`	Base solver classes (standalone)
`solver/solver_ilp.py`	SkyHOSTop MILP (SCIP-based)
`scripts/eval_baselines.py`	Runs MILP + all baselines
`scripts/plot_impact.py`	Generates figures from results
`results/mc_eval_results.csv`	Pre-computed results (720×4)
`requirements.txt`	Python dependencies

III. User Guide

We provide two reproduction paths: a **quick path** that regenerates figures from pre-computed results, and a **full path** that re-runs the entire evaluation from scratch.

A. Prerequisites

- **Python 3.10 or 3.11** is required. Python 3.12+ is not compatible due to dependency constraints.
- A virtual environment is recommended.

B. Quick Path: Regenerate Figures

This path requires only standard Python packages and takes under one minute. No MILP solver installation is needed.

```
pip install matplotlib pandas numpy
python scripts/plot_impact.py \
  --csv results/mc_eval_results.csv \
  --outdir figures/
```

This produces the two publication figures:

- `fig1_feasibility_vs_instances.pdf`
- `fig2_milp_vs_xron.pdf`

C. Full Path: Re-run Evaluation

To re-run the MILP and baseline evaluation from scratch, follow Steps 1–3 below.

1) Step 1 — Install Dependencies:

```
pip install -r requirements.txt
```

The MILP solver requires SCIP (version 9.0 was used in the paper) via the `pyscipopt` Python interface. The solver is fully self-contained and does not require any external frameworks.

2) Step 2 — Run the Evaluation: Run from the repository root directory:

```
python scripts/eval_baselines.py \
  --throughput data/mc_throughput_real.csv \
  --latency data/mc_latency_real.csv \
  --cost data/mc_cost.csv \
  --pairs "aws:us-east-1:azure:brazilsouth,
    aws:us-east-1:gcp:europe-west1,
    gcp:asia-northeast1:azure:eastus,
    azure:eastus:aws:eu-west-1,
    gcp:us-east1:aws:sa-east-1,
    azure:japaneast:gcp:us-west1" \
  --instance-limits "1,2,4" \
  --lambdas "1,2,3,5,8,10,12,15" \
  --budgets "100,200,300,500,1000" \
  --output results/mc_eval_results_new.csv
```

This runs 720 scenarios per baseline (8 stream rates $\times$ 5 latency budgets $\times$ 3 instance limits $\times$ 6 source-destination pairs) and takes approximately 20–30 minutes on a modern laptop.

3) Step 3 — Regenerate Figures:

```
python scripts/plot_impact.py \
  --csv results/mc_eval_results_new.csv \
  --outdir figures/
```

Dependency	Version
Python	3.10 or 3.11
NumPy	$\geq 1.21, < 2.0$
Pandas	$\geq 1.3, < 2.1$
Matplotlib	≥ 3.5
CVXPY	≥ 1.3
PySCIPOpt	≥ 4.0
SCIP	9.0 (recommended)

IV. Environment

A. Software Dependencies

The solver implementation is fully self-contained; no external cloud frameworks or accounts are needed.

B. Hardware Used in the Paper

- **Solver execution:** Apple M1 Pro, 16 GB RAM (any modern laptop with ≥8 GB RAM suffices).
- **Cloud measurements:** AWS m5.2xlarge, GCP n2-standard-8, Azure Standard_D8s_v3 instances across 18 regions. Cloud instances are **not** required for reproduction; all measurements are provided as CSV files.

V. Data & Methods

A. Data Origins

- **Throughput** (`mc_throughput_real.csv`): Measured via iperf3 with 16 parallel TCP streams and TCP buffer tuning on production cloud VMs. Each pair was measured twice and averaged.
- **Latency** (`mc_latency_real.csv`): Measured via ICMP ping (10 probes, averaged) between all 18×18 region pairs.
- **Egress cost** (`mc_cost.csv`): Collected from published provider pricing pages (AWS EC2, GCP VPC, Azure Bandwidth).

B. Topology

The evaluation uses 18 cloud regions (6 AWS, 6 GCP, 6 Azure) spanning North America, Europe, Asia-Pacific, and South America. Six cross-cloud source-destination pairs were selected with two pairs per provider combination (AWS↔GCP, AWS↔Azure, GCP↔Azure).

C. Parameter Sweep

Parameter	Values
Stream rate (λ)	{1, 2, 3, 5, 8, 10, 12, 15} Gbps
Latency budget	{100, 200, 300, 500, 1000} ms
VM limit ($N_{\max}$)	{1, 2, 4} per region
Source-dest pairs	6 cross-cloud pairs
Total scenarios	**720 per baseline**

VI. Assessment: Interpreting the Results

The pre-computed results file (`results/mc_eval_results.csv`) contains one row per scenario per baseline. Key columns:

- `baseline`: One of {MILP, XRON, Direct, Single-path}
- `feasible`: Whether a valid routing plan was found
- `cost_total_hr`: Total operational cost rate ($/hr)
- `solve_time_s`: Solver execution time (seconds)
- `batch_size_mb`: Optimized batch size (MB)

To verify the key claims from the paper:

1) **Feasibility:** Count `feasible=True` rows for each baseline across all 720 scenarios. Expected: MILP and XRON ≈82.6%, Single-path ≈55.1%, Direct ≈30.1%.
2) **Cost savings:** Filter to the 594 scenarios where both MILP and XRON are feasible. For each scenario, compute $(XRONcost - MILPcost)/XRONcost$ and average across all scenarios. Expected: ≈12.9%.
3) **Figures:** The regenerated figures should match the paper figures exactly when using the pre-computed results CSV.

VII. System Requirements

- **Quick path (figures only):** Any system with Python 3.10+. No special hardware or solver needed. Runs in under 1 minute.
- **Full path (re-run evaluation):** Python 3.10 or 3.11 with SCIP solver. 8 GB RAM recommended. No cloud accounts or GPUs required. Runs in 20–30 minutes.
- **Cloud measurements:** Provided as CSV files. Replicating measurements requires cloud accounts but is **not required** for verification.